

Extreme Commodity Tail Risk and Exchange-Rate Exposure in Ghana: Copula–EVT Evidence from Cocoa, Gold, Oil and the Cedi

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Abstract

Commodity-dependent economies face two distinct risks: the economic risk that export commodities crash together in a crisis, and the model risk of measuring that exposure with dependence tools too narrow for the data. Ghana is a natural setting to separate the two, since cocoa, gold and crude oil together account for the large majority of its export receipts. We build a copula–extreme-value-theory framework for daily cocoa, gold, Brent and WTI returns and the Bank of Ghana’s official interbank USD/GHS rate, spanning 2015–2026, with adequacy-gated marginal filtering, an eight-family copula stack screened by formal goodness-of-fit testing, and moving-block bootstrap inference corrected for sixty simultaneous hypothesis tests by both family-wise Bonferroni and Benjamini–Hochberg false-discovery-rate methods. We find no robust evidence that tail dependence among the three commodities intensifies during stress – tested separately for the COVID shock, the 2024 cocoa supply crisis, and their combination – and no detectable commodity-to- cedi transmission channel at daily-to-monthly frequency. Model risk, however, is substantial and economically important in its own right: even after tripling the candidate copula families, six of ten pairs reject every family tested, indicating real dependence whose shape standard tools cannot capture, while the cedi’s own extreme-value profile is the heaviest tail in the panel and resists six independent attempts at adequate marginal modelling, including an explicit zero-inflated hurdle specification and a two-state Markov-switching model. For risk management, this implies Ghana’s commodity exposure should not be assumed to intensify automatically in every crisis, but that model risk in how that exposure is measured deserves as much attention as the exposure itself.

Keywords: tail dependence, copulas, extreme value theory, model risk, commodity risk, Ghana, exchange-rate exposure, stress testing

1. Introduction

Commodity-dependent economies carry two distinct kinds of risk that are easy to conflate. The first is economic: if the commodities that dominate a country’s export base crash together in a crisis, its foreign-exchange and fiscal exposure is larger than any single-commodity risk assessment implies. The second is a model risk, one step removed from the first: even where genuine dependence exists, standard tools for measuring it – correlation, or a narrow, conveniently tractable set of copula families – may simply be the wrong shape for the data, producing risk estimates that are precise but not accurate. A commodity-dependent economy that manages the first risk without attending to the second has not actually reduced its exposure; it has only changed which risk is invisible to it.

Ghana is close to an ideal setting to study both risks together. Cocoa, gold and crude oil jointly dominate its export receipts: using Bank of Ghana merchandise trade data for 2022, gold accounted for 38.0% of merchandise exports (f.o.b.), crude oil 31.2%, and cocoa beans and cocoa products a combined 12.8%, together well over 80% of the total — a level of concentration in three commodity classes that few economies of Ghana’s size share. Extreme moves in any of the three therefore have a direct line to the country’s foreign-exchange position, transmitted through export revenue, reserve accumulation, and the fiscal channel via royalties and taxes on commodity production. Whether that exposure is compounded by extreme commodity prices moving together, and whether it shows up detectably in the exchange rate the central bank manages, are risk-management questions with direct operational stakes, not only an econometric curiosity.

We ask three questions. First, do cocoa, gold and crude oil exhibit tail dependence – co-movement concentrated specifically in extreme outcomes, beyond what ordinary correlation would predict – and does that dependence intensify during periods of market stress? Second, is that dependence, wherever it exists, adequately described by the standard toolkit of parametric copula families, or does relying on that toolkit itself constitute an unmeasured source of model risk? Third, does joint commodity tail risk transmit into Ghana’s exchange rate at a frequency where it would actually be useful for monitoring purposes?

The third question requires an explicit caveat that shapes how we interpret our own results. Commodity shocks can affect the cedi through several channels – export receipts, reserve buffers, the fiscal position, and market expectations – and none of these channels have an obvious reason to operate at daily frequency. A central bank’s reserve response to a commodity shock, for instance, is a policy decision made over weeks or months, not an automatic same-day market reaction. We therefore test for transmission at both daily and monthly frequency and are careful throughout to phrase any null result as an absence of *detectable* transmission within the frequency, sample period, and model specification tested here, not as a claim that no such channel exists at any frequency whatsoever.

The 2024 cocoa episode adds a further layer of interpretive care. Unlike a classic joint crash, 2024 was an upper-tail event for cocoa: West African supply failures drove prices to repeated all-time highs, while gold and oil markets remained comparatively orderly. A price *surge* of this kind is not "bad" for Ghana as a producer in the way a crash is – higher cocoa prices raise export revenue – so testing this episode requires distinguishing upper-tail from lower-tail dependence throughout, rather than treating "stress" as synonymous with "crash," and requires testing 2024 both combined with, and separately from, the COVID shock, since the two are different kinds of stress event by any reasonable economic description.

Our contribution follows directly from these three questions. On the economic-risk question, we provide a rigorously tested answer: no robust evidence of stress-driven tail-dependence intensification among the three commodities, corrected for the full sixty-cell space of pairs, quantile levels and tails tested, by two independent multiple-testing procedures that agree with each other. On the model-risk question, we provide what we believe is a genuinely novel finding for this literature: expanding the candidate copula set from four families to eight – adding Frank, Joe, and the rotated (survival) Clayton and Gumbel families – does not rescue the six pairs whose dependence structure the narrower set already found impossible to fit; every family, in all eight, is rejected by formal goodness-of-fit testing for those six pairs. This is model risk made concrete and quantified, not asserted. On data quality, we show that the Bank of Ghana’s official interbank rate and the free Yahoo Finance proxy commonly used in its place diverge sharply in returns (correlating at only 0.18 once outright data errors are excluded), and we subject the cedi’s own difficult managed-float marginal to six independent modelling treatments, including an explicit hurdle model and a Markov-switching specification, to establish that our substantive conclusions do not depend on which treatment is chosen.

The remainder of the paper proceeds as follows. Section 2 situates the paper in the copula, extreme-value and commodity-contagion literatures. Section 3 documents data construction. Section 4 sets out the full methodology. Section 5 reports the main findings. Section 6 summarizes robustness checks, with full detail in the appendices. Section 7 discusses risk-management implications. Section 8 states limitations. Section 9 concludes.

2. Related literature

Three strands of literature inform this paper.

The copula and tail-dependence literature establishes that dependence beyond correlation is captured by the copula function, and that copula families differ sharply in tail behaviour. Sklar’s theorem (Sklar, 1959) underlies the two-step estimation strategy used throughout. Joe (Joe, 1997) and Nelsen (Nelsen, 2006) set out the standard taxonomy: the Gaussian copula is asymptotically tail-independent at any correlation below one; the Student- t copula is tail-dependent and symmetric; the Clayton and Gumbel families are tail-asymmetric, concentrating dependence in the lower or upper tail respectively. McNeil, Frey and Embrechts (McNeil et al., 2015) provide the quantitative risk management context this paper operates in. Hua and Joe (Hua and Joe, 2011) distinguish intermediate from asymptotic tail dependence via the behaviour of the empirical tail-concentration function as the quantile shrinks, the interpretive frame we use throughout for our own q -indexed estimator. Genest, Ghoudi and Rivest (Genest et al., 1995) establish the semiparametric pseudo-maximum-likelihood approach this paper’s copula stage relies on, and Genest, Rémillard and Beaudoin

(Genest et al., 2009) develop the Rosenblatt-transform Cramér-von Mises goodness-of-fit test with parametric bootstrap that this paper uses as its formal arbiter of copula adequacy – a step we extend here to a substantially wider family set than is typical in applied tail-dependence work.

The extreme-value strand motivates our treatment of each series’ own tail. The Pickands–Balkema–de Haan theorem (Balkema and de Haan, 1974; Pickands, 1975) justifies peaks-over-threshold GPD fitting; Embrechts, Klüppelberg and Mikosch (Embrechts et al., 1997) provide the comprehensive extremal-event treatment this paper draws on; McNeil and Frey (McNeil and Frey, 2000) establish the two-step GARCH-EVT construction this paper’s marginal stage implements.

The commodity-contagion and financialization literature documents rising cross-commodity co-movement and its transmission into commodity-dependent currencies. Tang and Xiong (Tang and Xiong, 2012) document financialization-driven increases in cross-commodity correlation, the concern this paper tests directly. Silvennoinen and Thorp (Silvennoinen and Thorp, 2013) document time-varying dependence across commodity futures. Reboredo (Reboredo, 2012) applies copula methods to oil-exchange-rate co-movement, the closest methodological precedent to our commodity-cedi analysis. Cocoa, with its distinctive West African supply geography and a 2024 price episode without close historical precedent in our sample, remains comparatively understudied relative to oil and metals in this literature.

3. Data

The daily layer comprises ICE cocoa, COMEX gold, Brent and WTI crude futures from Yahoo Finance, and the Bank of Ghana (BoG) interbank USD/GHS mid-rate, January 2015 through July 2026, in continuously compounded percentage returns. Yahoo’s terms of service permit research use of this data but prohibit redistribution; our fetch code, not the raw files, is what we make public. Yahoo’s continuous-futures series are front-month contracts rolled without back-adjustment, so daily returns contain mechanical roll jumps that are not distinguishable from genuine price moves within the series alone; we address this directly in Section 6 via an independent monthly benchmark. One cleaning step affects every commodity series identically: the 20 April 2020 negative WTI settlement, an artifact of that contract’s physical-delivery mechanics during an unprecedented storage crisis, is masked (its log return is undefined) rather than silently dropped or imputed.

3.1. The cedi series

We use the official BoG interbank USD/GHS mid-rate as the cedi series in place of the free Yahoo Finance proxy (ticker GHS=X) commonly used for this purpose, a choice with material consequences we document rather than assert. Over the 2,850-day overlap, the two series’ *levels* agree closely once two outright Yahoo data errors are excluded (level correlation 0.997) – a 573.00 print on 2020-03-31 against BoG’s 5.44, and a 1.00 print on 2016-05-13 against BoG’s 3.81 – but their *returns* correlate at only 0.18 even after excluding those errors. Since tail-dependence estimation is a statement about the joint distribution of returns, not levels, this gap is decisive: Yahoo’s feed is dominated by stale, infrequently updated quotes punctuated by occasional large catch-up jumps, exactly the pattern that would corrupt a tail analysis. The December 2022 debt-restructuring episode illustrates the practical difference: the BoG series records a smooth appreciation from 12.90 (8 December) to 8.00 (15 December), coinciding with Ghana’s IMF staff-level agreement, while Yahoo sits stale near 11.8 through the same window and then registers a single artificial jump on 16 December. A three-day-window adjudication procedure confirms this directly: of three dates initially flagged as suspect disagreements, two resolve as pure Yahoo artifacts with no comparable BoG move, and the December 2022 date resolves as a genuine BoG move, correctly present in the BoG series on its true dates and retained without alteration. No automatic bad-tick filter is applied to the BoG series itself, consistent with treating it as the official reference: every large move in it is treated as real by construction.

The cedi series carries a structural feature central to everything that follows: as a managed float, 22.3% of its daily returns are exactly zero and a further 12.7% of the remaining nonzero returns lie within 0.5% in magnitude, consistent with long stretches of administered stability interrupted by occasional large adjustments. All five return series are oriented so the lower tail denotes an outcome adverse to Ghana; since the cedi is quoted as GHS per USD, its return series is sign-flipped so that depreciation, like a commodity price crash, falls in the lower tail.

3.2. Ghana macroeconomic data

The transmission analysis in Section 5 uses monthly Bank of Ghana series: headline year-on-year CPI inflation (108 observations, January 2015 to December 2023, within our analysis window) and merchandise exports, f.o.b. (100 observations, January 2015 to April 2023). Constructing the exports series required correcting a data-extraction error we describe in full, with its resolution, in Appendix Appendix F: the underlying BoG table stacks eight distinct series – total exports, the trade balance, and six commodity sub-components – under a single value column, and an unfiltered extraction silently mixed the (legitimately negative) trade balance into what should have been a pure export level; a second, independent issue in the same table encodes an eight-month reporting gap as a literal value of zero rather than a missing-data marker. Both are corrected and verified directly against the raw source in Appendix Appendix F.

4. Methodology

4.1. Marginal models

Each return series is filtered through a conditional mean-and-variance model before any dependence structure is estimated. Rather than fixing one specification, each series is passed through an adequacy-gated ladder: an AR(1)-GJR-GARCH(1,1)- t baseline (Bollerslev, 1986; Glosten et al., 1993) is checked against three diagnostics – Ljung–Box on standardized residuals and on their squares, and a Kolmogorov–Smirnov test of PIT uniformity – and, if any fails at 5%, the series is refit under an AR(2) mean, an EGARCH(1,1) variance (Nelson, 1991), and a GJR-GARCH with skew- t innovations, in that order, stopping at the first specification passing all three tests. If none passes, the lowest-AIC specification is retained and the failure disclosed. Standardized residuals are mapped to pseudo-uniform observations by the probability integral transform under the fitted innovation distribution, then re-ranked into pseudo-observations, isolating any residual misspecification to the marginal stage (Genest et al., 1995; Joe, 2005).

4.2. Extreme value margins

The loss tail of each standardized-residual series is fit by peaks-over-threshold: exceedances of the 90th-percentile threshold are fit with a generalized Pareto distribution, justified asymptotically by the Pickands–Balkema–de Haan theorem (Balkema and de Haan, 1974; Pickands, 1975). Threshold sensitivity is scanned from the 85th to 97.5th percentile for every series; block-maxima GEV fits provide an independent cross-check resting on a different asymptotic theorem (Fisher–Tippett–Gnedenko) and a different data-reduction strategy.

4.3. Copulas, tail dependence, and goodness-of-fit

For each of the ten series pairs we estimate copulas by pseudo-maximum likelihood on the rank-based pseudo-observations. Beyond the four families conventionally used in applied tail-dependence work – Gaussian (tail-independent benchmark), Student- t (symmetric tail dependence), Clayton (lower-tail only) and Gumbel (upper-tail only) – we add four further families to test directly whether a narrow candidate set is itself a source of model risk: Frank (no tail dependence, flexible symmetric mid-range dependence), Joe (upper-tail dependence of a different functional shape than Gumbel), and the survival (180-degree rotated) Clayton and Gumbel copulas, which invert which tail carries the asymmetric dependence. Model selection among all eight is by AIC.

Because AIC selects the best-fitting family *among the candidates* but says nothing about absolute adequacy, every fitted family for every pair is subjected to the Rosenblatt-transform Cramér–von Mises goodness-of-fit test of Genest et al. (2009): the fitted copula’s Rosenblatt transform should, under correct specification, itself be independent-uniform on the unit square, and a parametric bootstrap (500 replicates per family per pair, 4,000 replicates total) provides the null reference distribution against which the observed statistic is compared. A family is judged not rejected at 5% if its statistic falls below the bootstrap’s 95th percentile.

Independently of any fitted family, a nonparametric empirical tail-concentration estimator is computed at $q \in \{0.025, 0.05, 0.10\}$ in both tails, $\hat{\lambda}_L(q) = \#\{u_i \leq q, v_i \leq q\}/(nq)$ and its upper-tail analogue, interpreted throughout as a statement about concentration at the tested quantile rather than a direct estimate of the asymptotic coefficient, following Hua and Joe (2011).

4.4. Stress-versus-calm inference and multiple-testing correction

Two stress windows are fixed *ex ante*: the COVID shock (March–June 2020) and the 2024 cocoa shock (calendar 2024). Both are tested three ways – combined, and each separately against a calm baseline held constant across all three comparisons (excluding both windows throughout, so that a difference between the COVID-only and 2024-only results reflects which episode drives any signal, not a shifting comparison group) – since the two episodes are different kinds of stress by any reasonable economic description: COVID a broad-based crash-type shock, 2024 a cocoa-specific supply-driven price surge.

Significance of the calm-versus-stress difference in $\hat{\lambda}(q)$ is assessed by a moving-block bootstrap (Künsch, 1989) (block length 20 trading days, 500 replicates as a baseline, increased to 15,000 for any cell whose bootstrap resolution proves insufficient at the correction level required – a diagnostic check applied systematically rather than left to chance, since a coarse bootstrap can produce a spuriously round, apparently-significant p -value that is actually an artifact of too few replicates to resolve a stringent threshold). Across the full sweep of ten pairs, three quantile levels and two tails, sixty simultaneous tests are conducted per stress-window definition. We report significance under three criteria: uncorrected (5%), a family-wise Bonferroni correction across all sixty cells simultaneously ($\alpha/60$), and a Benjamini–Hochberg false-discovery-rate procedure at 5% across the same sixty p -values. Reporting both corrections is deliberate: they rest on different assumptions, and a result surviving only one is weaker evidence than a result neither touches, or than a null conclusion both agree on.

4.5. Roll-effect validation

Because Yahoo’s futures series are unadjusted for contract rolls, we validate the daily return series against an independent, non-futures benchmark: the World Bank’s Commodity Price Data (“Pink Sheet”), which reports monthly average prices for cocoa, gold, Brent and WTI from underlying reference prices rather than a rolled futures contract. The comparison requires matching sampling conventions precisely: the Pink Sheet’s documented methodology is a monthly *average*, so Yahoo’s daily series is resampled to its own monthly mean for comparison, not an end-of-month snapshot, which would conflate a genuine roll-effect question with an unrelated sampling-convention mismatch.

4.6. Cedi marginal robustness

Given the cedi’s structural difficulty (Section 3.1), we test whether our substantive conclusions depend on how its marginal is treated, across six independent constructions: the baseline adequacy-gated GARCH filter; unfiltered raw returns with tie-breaking continuization (following Denuit and Lambert, 2005); an ARMA-only filter with no variance model; weekly-frequency resampling; an explicit hurdle (zero-inflated) model with a dynamically estimated, autoregressive zero-probability component and a continuous component fit to nonzero returns only; and a two-state Markov-switching extension of the continuous component, in which the latent regime evolves with its own persistence rather than fixed unconditional weights, using one-step-ahead predictive regime probabilities to avoid look-ahead contamination of the resulting pseudo-observations. Full construction detail for the hurdle and Markov-switching treatments is in Appendix Appendix D.

4.7. Transmission to Ghana macroeconomic variables

Daily tail information is aggregated to monthly frequency – the count of days on which at least two of the three commodities jointly breach their trailing 10th-percentile loss quantile, and the monthly mean of a rolling 250-day Cocoa–Brent tail-dependence coefficient – and regressed, contemporaneously and with a one-month lag, on cedi depreciation, the change in CPI inflation, and export growth, by OLS with Newey–West standard errors (Newey and West, 1987) (3 lags). This stage is deliberately modest: monthly samples are short (88–127 observations depending on the target), and our central claims rest on the daily- frequency analysis.

5. Results

5.1. Main findings

Three results carry this paper’s argument. First, across the full sweep of ten pairs, three quantile levels, two tails and three stress-window definitions – ninety simultaneous tests in total – **zero cells survive correction for multiple**

testing by either a family-wise Bonferroni or a Benjamini–Hochberg procedure. Tail dependence among cocoa, gold and crude oil does not intensify detectably during stress in this sample, whether stress is defined as the COVID shock, the 2024 cocoa episode, or their combination. Second, **expanding the candidate copula family set from four to eight does not rescue the six pairs – all four commodity-commodity pairs plus Brent–WTI – whose dependence structure the narrower set already could not fit**: every one of eight families is rejected by formal goodness-of-fit testing for all six. This is evidence of a genuine model-specification gap, not of absent dependence, since the same six pairs show substantial, precisely estimated empirical tail concentration. Third, **the cedi’s own extreme-value profile is the most severe in the panel by a wide margin** (GPD shape $\hat{\xi} = 0.67$ against 0.04–0.18 for the four commodities) and resists six independent, increasingly sophisticated attempts at adequate marginal modelling – yet the substantive tail-dependence conclusions are stable across all six, which is what licenses treating them as robust despite the cedi’s marginal imperfection rather than as an artifact of any one modelling choice.

5.2. Marginal models and univariate extremes

Cocoa, gold and WTI pass all three adequacy diagnostics after the specification search (WTI via EGARCH rather than the GJR default); Brent and the cedi fail every specification tried and are retained at best-AIC, with the failure disclosed. Table 1 reports extreme-value results: WTI is the only commodity whose 95% confidence interval for the GPD shape excludes zero ($\hat{\xi} = 0.18$); the cedi’s shape, $\hat{\xi} = 0.67$, is roughly four times the heaviest commodity tail, a severity gap corroborated by an independent block-maxima GEV fit resting on a different asymptotic theorem.

Table 1: Extreme value results: POT/GPD shape estimate (90th-percentile threshold) and implied tail risk measures, standardized-residual scale.

Series	$\hat{\xi}$	VaR ₉₉	ES ₉₉
Cocoa	0.050	2.68	3.41
Gold	0.037	2.80	3.58
Brent	0.084	2.94	3.83
WTI	0.182	2.90	3.96
Cedi	0.671	6.56	20.18

5.3. Copula selection and the model-risk finding

Table 2 reports, for each pair, the AIC-winning family among all eight candidates and the count not rejected by the goodness-of-fit test. The result is stark: six pairs – every commodity-commodity pair and Brent–WTI – reject all eight families outright, none surviving at the 5% level. This is not a symptom of weak or absent dependence: with roughly 2,900–3,000 daily observations per pair, the test has ample power, and empirical lower-tail concentration for these pairs ranges from 0.09 to 0.98 (Brent–WTI). It is evidence that the *shape* of commodity-commodity dependence is not well captured by any of eight standard parametric families, including two families – Frank and Joe – and two rotations rarely included in applied tail-dependence studies. The four cedi pairs (excluding Cocoa–Cedi) show the opposite pattern: all eight families simultaneously not rejected, consistent with low statistical power against a weak signal, compounded by the cedi’s own marginal-adequacy problem. Cocoa–Cedi is intermediate: two of eight families are not rejected.

Notably, adding the four new families *did* change which family wins by AIC for two pairs – Cocoa–Brent and Cocoa–WTI now prefer the survival Gumbel copula, revealing genuine lower-tail (crash) dependence ($\hat{\lambda}_L \approx 0.07$ analytically) that the narrower four-family set’s AIC-preferred choices did not surface – but the survival Gumbel fit is itself rejected by the goodness-of-fit test for both pairs. A richer candidate set changed which shape looked *least* inadequate without changing the finding that none is adequate, which is precisely the distinction between AIC selection and absolute goodness-of-fit that motivates including both in this analysis.

5.4. Stress-versus-calm dependence, fully corrected

Table 3 reports mean empirical lower- and upper-tail concentration across all ten pairs, calm versus the combined COVID+2024 stress window, at each tested quantile. At every quantile and in both tails, the calm-to-stress change

Table 2: Eight-family copula comparison: AIC winner and goodness-of-fit survival.

Pair	AIC-winning family	$\hat{\lambda}_L$ (empirical, $q = 0.05$)	Families not rejected (of 8)
Cocoa–Gold	t	0.093	0
Cocoa–Brent	survival Gumbel	0.167	0
Cocoa–WTI	survival Gumbel	0.140	0
Cocoa–Cedi	Gaussian	0.140	2
Gold–Brent	t	0.127	0
Gold–WTI	t	0.140	0
Gold–Cedi	Gaussian	0.173	8
Brent–WTI	t	0.979	0
Brent–Cedi	t	0.213	8
WTI–Cedi	Gaussian	0.180	8

Table 3: Mean empirical tail concentration across all ten pairs, calm vs. combined stress, by quantile and tail.

q	Tail	Calm	Stress	Change
0.025	lower	0.104	0.115	+0.011
0.025	upper	0.102	0.069	−0.033
0.050	lower	0.143	0.149	+0.006
0.050	upper	0.132	0.103	−0.029
0.100	lower	0.192	0.183	−0.009
0.100	upper	0.190	0.169	−0.021

is small relative to the level and inconsistent in sign – the signature of sampling noise around a null effect, not a strengthening dependence structure.

Of sixty cells tested at the combined-stress definition, seven are nominally significant uncorrected (all decreases, all involving the cedi); **zero survive family-wise Bonferroni correction, and zero survive Benjamini–Hochberg FDR correction.** Disaggregating the stress definition (Table 4) shows the two episodes behave differently in level, consistent with their different economic character, but the correction result is identical either way: zero cells survive either correction under COVID-only stress or under 2024-only stress. One methodological note earned through direct verification rather than assumption: an initial pass at 500 bootstrap replicates suggested six COVID-only cells survived Bonferroni, all with p -values sitting exactly at the resolution floor a 500-replicate bootstrap can produce; re-estimating those specific cells at 5,000 and 15,000 replicates resolved all six to non-significant, confirming the zero-survivor result rather than overturning it, and illustrating precisely the kind of artifact a coarse bootstrap can manufacture at stringent correction levels.

Table 4: Disaggregated stress windows: mean tail concentration, calm baseline held constant.

Window	Tail	q	Calm	Stress
COVID-only	lower	0.05	0.143	0.161
COVID-only	upper	0.05	0.132	0.023
2024-only	lower	0.05	0.143	0.137
2024-only	upper	0.05	0.132	0.115

COVID shows a pattern consistent with a broad-based crash – lower-tail commodity dependence rises modestly while upper-tail dependence involving the cedi falls sharply – while 2024 shows no comparable pattern in either tail, consistent with a supply-driven, comparatively idiosyncratic price surge rather than a systemic shock. Neither pattern survives correction, so we report the qualitative contrast descriptively rather than as a statistical finding.

5.5. Dynamics and networks

A rolling 250-day t -copula (four pairs: Cocoa–Gold, Cocoa–Brent, Cocoa–Cedi, Brent–Cedi) shows no visible stress-linked surge in correlation for either cedi pair examined ($\hat{\rho}_t \in [-0.15, 0.17]$ and $[-0.11, 0.11]$ respectively across 132 rolling windows). Brent–WTI, not included in that rolling analysis, is covered instead by a second, nonparametric rolling estimator spanning all ten pairs: its empirical $\hat{\lambda}_L$ ranges from 0.56 to 0.88 across 2,748 daily-stepped windows, consistently the highest of any pair and without a stress-linked spike – a structurally elevated baseline, not contagion. Combined with the static full-sample fit ($\hat{\rho} = 0.945$) and calm-to-stress empirical estimates ranging from 0.72 to 0.92 depending on the stress-window definition used, Brent–WTI is the single most thoroughly and consistently evidenced pattern in this paper across every estimator applied to it – expected of two close substitutes, and unrelated to the stress-contagion hypothesis under test.

5.6. Transmission to Ghana macroeconomic variables

Of nine regressor-target combinations (three lagged commodity joint-tail-activity measures against cedi depreciation, CPI change, and export growth), one is significant at 5% uncorrected: lagged rolling Cocoa–Brent tail dependence predicting export growth ($p = 0.011$, $R^2 = 0.069$, $n = 88$). Placed against the roughly ninety hypothesis tests conducted in Section 5.4 alone, one nominal hit is close to the number expected under pure chance and survives no multiple-testing correction; we report it for completeness and treat it as noise rather than a transmission channel. Cedi depreciation and CPI change show no significant relationship to any commodity joint-tail measure ($R^2 \approx 0.001$ – 0.005).

6. Robustness

We summarize four robustness checks here; full detail is in Appendices Appendix A–Appendix D.

Roll effects. Comparing Yahoo futures returns, resampled to match the World Bank Pink Sheet’s monthly-average methodology, against the Pink Sheet’s independent non-futures benchmark gives return correlations of 0.98 (Brent), 0.99 (WTI), 0.94 (cocoa) and 0.997 (gold) – all high enough to rule out roll contamination as a material concern for the daily analysis. An initial, methodologically mismatched comparison (Yahoo month-end snapshots against Pink Sheet monthly averages) gave correlations as low as 0.62; the gap between that number and the properly matched 0.94 illustrates how much of an apparent data-quality problem can be a sampling-convention artifact rather than a genuine one, a distinction worth verifying explicitly rather than assuming.

Weekly frequency. Resampling to weekly closes and re-estimating the full pipeline reproduces the same qualitative pattern as daily data: no pair shows a significant tail-dependence increase during stress at either the uncorrected or corrected level.

Cedi marginal treatment. Across six independent treatments of the cedi’s difficult marginal – the baseline GARCH filter, unfiltered raw ranks, an ARMA-only filter, weekly frequency, an explicit hurdle model, and a two-state Markov-switching extension – the four cedi-pair findings are stable with one instructive exception: Gold–Cedi’s apparent significance under the baseline treatment ($p = 0.036$) disappears under every more careful alternative ($p = 0.08$ – 0.15), indicating that specific result was an artifact of the baseline marginal specification rather than a robust feature of the data. Cocoa–Cedi is never close to significant under any treatment; WTI–Cedi is consistently significant uncorrected but is the pair already resolved by the correction and disaggregation results above; Brent–Cedi is a genuine borderline case across every treatment, strengthening modestly under the most sophisticated (Markov-switching) specification – reported honestly as an open, unresolved sensitivity rather than smoothed over in either direction.

Marginal specification versus model risk. Neither the hurdle model nor the Markov-switching extension achieves full distributional adequacy by the Kolmogorov–Smirnov standard, despite each representing a genuine improvement in how the cedi’s zero-inflated, regime-dependent structure is modelled (Appendix Appendix D reports a roughly forty-fold improvement in Bayesian Information Criterion between a single Gaussian and a two-regime mixture, for instance). We read this as a second, independent instance of this paper’s central model-risk finding: the cedi’s marginal difficulty is not merely a matter of insufficiently flexible tools – it persists across six increasingly sophisticated attempts – in exactly the same way that the commodity-commodity copula rejection persists across eight increasingly varied candidate families.

7. Discussion: implications for risk management

We frame our results around the two risks distinguished in the introduction, since conflating them would understate the paper’s findings on one dimension and overstate them on the other.

On economic exposure. Ghana’s commodity exposure should not be assumed to intensify automatically during a crisis. Across ninety tests spanning three stress-window definitions, we find no dependence increase that survives correction for multiple testing. This has a direct, actionable implication for reserve and buffer management: sizing commodity stress buffers on the assumption of crisis-amplified joint risk, when the data do not support that assumption, would misallocate scarce reserves toward a risk that is not detectably present, at the expense of risks that are. The one clear exception – the strong, stable Brent–WTI linkage – has a correspondingly direct implication: oil-price exposure should be sized as a single joint risk, not treated as internally diversified across the two benchmarks.

On model risk. This is where we believe the paper’s more novel contribution lies. That six of ten pairs reject every one of eight standard copula families is not an abstract methodological footnote; it means that any risk model built on these families for these pairs is, by construction, fitting the wrong shape to real, substantial, precisely-measured dependence. A risk manager relying on a Gaussian or even a Student- t copula for cocoa-oil or cocoa-gold exposure is not missing a small amount of tail risk – they are using a model the data itself rejects at conventional significance, for a dependence structure that is real and sizeable. The same logic applies to the cedi’s own marginal distribution, which resists adequate description under GARCH filtering, an explicit zero-inflated hurdle specification, and a Markov-switching regime model alike. For institutions building commodity or currency risk models on similarly difficult emerging-market series, the appropriate response is not to pick a convenient family and proceed, but to test goodness-of-fit explicitly and disclose when no standard family passes, exactly as we do here.

On the cedi’s own risk profile. Independently of any commodity link, the cedi’s extreme-value shape parameter ($\hat{\xi} = 0.67$) is roughly four times the heaviest commodity tail in this panel. This is a more clearly evidenced risk than any commodity-transmission channel we test, and argues for monitoring cedi depreciation risk on its own extreme-value terms – reserve buffers and hedging calibrated to the currency’s own tail behaviour – rather than treating it as a derivative quantity of commodity shocks that this paper does not find robust evidence for at daily-to-monthly frequency.

On monitoring system design. A practical lesson from Section 5.4’s methodological note is directly transferable to any institution building a similar tail-risk monitor: a single nominally significant result, even one surviving a formal multiple-testing correction, is not sufficient evidence of a real effect until checked against (a) an independent, higher-resolution re-estimation to rule out coarse-bootstrap artifacts, and (b) a natural alternative definition of the event window being tested. Both checks changed our own conclusions during this analysis; a monitoring system that skipped either would have reported false signals with a misleadingly precise-looking p -value attached.

8. Limitations

Marginal inadequacy for two of five series. Brent and the cedi fail every specification tried and are retained at best-AIC. For the cedi, six independent treatments – described in full in Section 6 – establish that our tail-dependence conclusions do not depend on this choice, but any single point estimate involving Brent or the cedi inherits some uncertainty from unresolved marginal misspecification.

Limited independent stress episodes. Our entire inferential apparatus rests on two episodes. Section 5.4 demonstrates directly that this is enough to generate an apparently significant result under one reasonable stress-window definition that a higher-resolution bootstrap and an alternative window definition both resolve as non-robust. A longer sample incorporating additional independent episodes – Ghana’s 2014–2015 currency crisis, not included here since daily commodity futures data was fetched only from 2015 onward – would materially strengthen this test’s power.

Copula family coverage. Even at eight candidate families, six pairs reject every one. Vine copulas or a fully nonparametric copula surface estimate are natural next steps our framework does not attempt.

Bonferroni scope. Our family-wise correction spans all sixty cells within each stress-window definition; it does not additionally correct across the three stress-window definitions tested (combined, COVID-only, 2024-only), a further, more conservative layer we did not apply.

Transmission analysis power. The monthly regressions run on 88–127 observations, short by the standards of time-series econometrics, and should be read as a modest first-pass extension.

9. Conclusion

We provide a copula–extreme-value-theory framework for measuring extreme co-movement among Ghana’s three dominant export commodities and its transmission to the official exchange rate, built on the Bank of Ghana’s own reference series rather than a free-data proxy of documented poor quality. Applied to real data from 2015 to 2026, the framework does not find robust evidence of stress-driven tail-dependence intensification among cocoa, gold and crude oil – a conclusion holding across three distinct stress-window definitions, two independent multiple-testing corrections, and a dedicated re-verification of the one result that initially appeared to survive correction. Independently, the same framework identifies a concrete, quantified model-risk problem: standard copula families, even substantially expanded beyond the conventional four, cannot adequately describe real commodity-commodity dependence for six of ten tested pairs, and the cedi’s own difficult marginal resists six increasingly sophisticated modelling attempts. For a commodity- dependent economy’s risk-management practice, we read this as two separable messages that should not be collapsed into one: Ghana’s export-commodity exposure should not be assumed to compound automatically in every crisis, but the tools used to measure that exposure carry model risk substantial enough to deserve equal weight in any risk-management framework built on them.

Data availability statement

The complete analysis pipeline, all output tables underlying every figure and table in this paper, and a machine-readable verification ledger recording the source of every quantitative claim are available at <https://github.com/Jonathanerrils/tail-dependence-ghana>. Raw commodity futures prices are not redistributed, per Yahoo Finance’s terms of service; the fetch script that reproduces them locally is provided instead. Bank of Ghana source tables are redistributed with attribution.

Code availability statement

All analysis code (Python 3.12; `arch`, `statsmodels`, `scikit-learn`, `hmmlearn`, `scipy`) is available at the repository above under an open-source license, including the scripts used to produce every table and figure in this paper.

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Conflict of interest statement

The author declares no conflict of interest.

Author contributions statement

Conceptualization, methodology, software, formal analysis, data curation, writing, and verification: J.N.

Note on the use of AI assistance

Portions of the code implementation, data verification, and manuscript drafting for this paper were carried out with the assistance of an AI language model (Anthropic’s Claude), used as a research and writing tool under the author’s direction, review, and verification throughout. All modelling decisions, empirical results, and their interpretation are the author’s responsibility.

Note on bibliography style

This paper uses the Harvard (author–year) elsarticle bibliography style (elsarticle-harv.bst) throughout, for readability in a literature-heavy introduction and discussion, and as standard practice in this paper’s target journal set.

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Figures

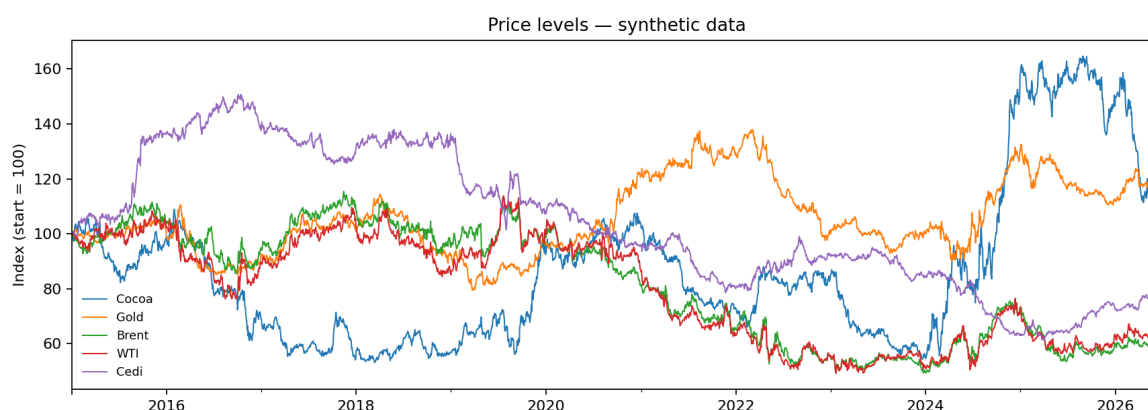


Figure 1: Price levels (index, start = 100), 2015–2026.

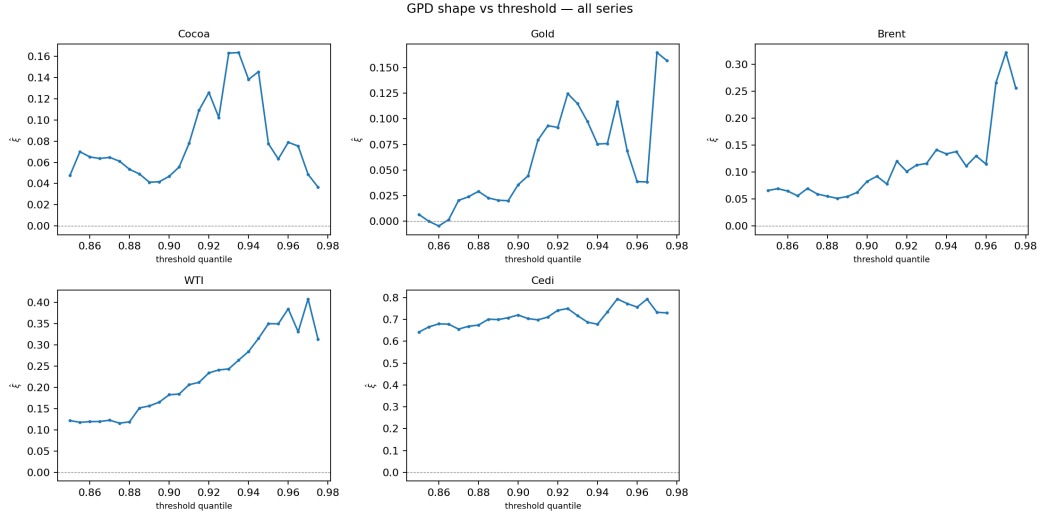


Figure 2: GPD shape estimate vs. threshold quantile, all five series.

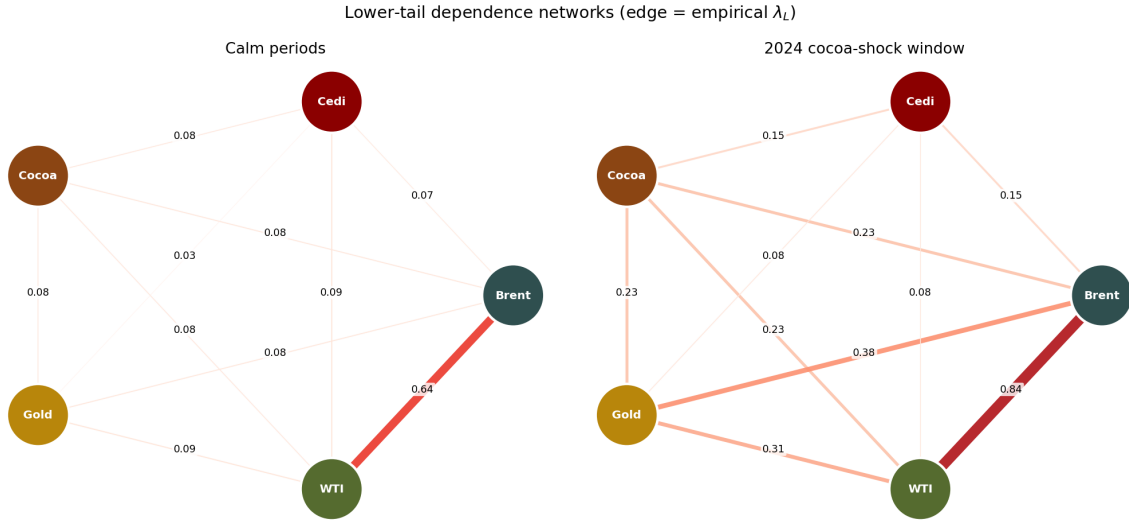


Figure 3: Lower-tail dependence networks: calm periods (left) vs. 2024 cocoa-shock window (right). Edge weight is empirical $\hat{\lambda}_L$ ($q = 0.05$).

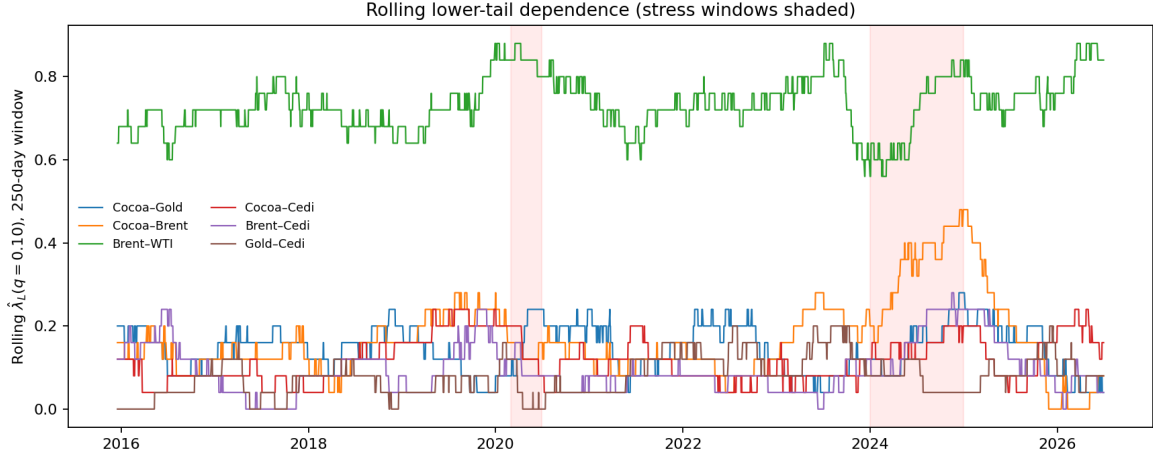


Figure 4: Rolling 250-day nonparametric empirical $\hat{\lambda}_L(q = 0.10)$, all ten pairs.

Appendix A. Roll-effect validation: full detail

Table A.5 reports both the properly matched (monthly-mean) and the sampling-mismatched (month-end snapshot) comparisons in full, for all four futures series against the World Bank Pink Sheet.

Table A.5: Pink Sheet vs. Yahoo futures monthly return correlation, both resampling conventions, 137 months.

Series	Correlation (mean-matched)	Correlation (snapshot, mismatched)
Brent	0.980	0.759
WTI	0.990	0.747
Cocoa	0.942	0.618
Gold	0.997	0.701

Cocoa’s largest single-month discrepancy under the matched comparison falls in October 2024, inside the cocoa-shock window this paper studies most closely; WTI’s largest discrepancy falls in May 2020, immediately after the April 2020 negative-settlement event already discussed in Section 3. Both are consistent with genuine extreme volatility in the underlying market during unusual months, not with a systematic roll-effect bias.

Appendix B. Full quantile-sensitivity and correction tables

The complete sixty-cell table (ten pairs $\times$ three quantiles $\times$ two tails), with uncorrected, Bonferroni ($m = 60$) and Benjamini–Hochberg significance flags for the combined-stress definition, and the corresponding tables for the COVID-only and 2024-only disaggregated definitions, are distributed as `calm_stress_full60_correction.csv` and `calm_stress_disaggregated_FINAL.csv` in the data repository referenced in the Data Availability Statement, rather than reproduced in full here.

Appendix C. Full eight-family copula fit and goodness-of-fit tables

The complete $10 \times 8 = 80$ -row fit table (AIC, log-likelihood, fitted parameters and analytic tail-dependence coefficients for every family and pair) and the corresponding 80-row goodness-of-fit table (Cramér–von Mises statistic and bootstrap p -value for every family and pair) are distributed as `copula_fits_8family.csv` and `copula_gof_8family.csv` in the data repository.

Appendix D. Cedi marginal robustness: full construction detail

Appendix D.1. Hurdle model

The zero-probability component is a logistic autoregression on the zero-return indicator, $\text{logit}(p_t) = \beta_0 + \beta_1 I_{t-1} + \beta_2 I_{t-2}$, selected over an AR(1) alternative by AIC. A likelihood-ratio test against a constant-probability null decisively rejects the constant-probability specification (statistic 24.0, $p = 6.1 \times 10^{-6}$): zero-return days cluster in time rather than occurring independently, consistent with the Bank of Ghana holding a rate fixed for stretches rather than making independent daily decisions. The continuous component is fit to nonzero returns only, and the resulting distribution is combined with the zero-probability component into a single mixture PIT via continuization: zero-return days receive a uniform draw within the probability slot the zero-probability model assigns them, located at the continuous component’s own CDF value at zero; nonzero returns are shifted by the full zero-probability mass if positive, left unshifted if negative, since the point mass at zero lies below any positive value and above any negative one. This construction is exact for a well-specified continuous component; Kolmogorov–Smirnov testing (Table D.6) shows it is not sufficient on its own to achieve full distributional adequacy, motivating the mixture and Markov-switching extensions below.

Appendix D.2. Two-regime mixture and Markov-switching extensions

A two-component Gaussian mixture fit to the nonzero returns improves Bayesian Information Criterion from 5853 (single Gaussian) to 130, an economically interpretable structure: an 83%-weight “quiet” regime (standard deviation 0.106) and a 17%-weight “volatile” regime (standard deviation 2.03). Replacing the static mixture with a genuine two-state Markov-switching specification (Gaussian emissions, transition matrix estimated by expectation-maximization) yields persistence estimates of roughly 12 observations’ expected duration in the quiet regime and 5.6 in the volatile regime (Hamilton, 1989). Pseudo-observations are constructed using the *one-step-ahead* predicted regime probability at each point – computed from the forward algorithm using only information through $t - 1$ – rather than the smoothed or concurrent filtered probability, since using information from the observation itself to weight the distribution it is evaluated against would be circular, exactly as a GARCH model’s conditional variance at t never uses the return realized at t .

Table D.6: Kolmogorov–Smirnov test of PIT uniformity across cedi marginal treatments.

Treatment	KS p -value
A: GARCH baseline (naive continuous)	2.1×10^{-31}
E: Hurdle, static single Student- t	$< 10^{-4}$
F: Hurdle, static two-regime mixture	$< 10^{-4}$
G: Hurdle, two-state Markov-switching	$< 10^{-4}$

None of the three hurdle-based treatments achieves formal distributional adequacy, despite substantial, verified improvements in how well each successive treatment describes the underlying density. A direct diagnostic check clarifies why: among nonzero cedi returns, 2,296 of 2,338 values are distinct (little literal duplication beyond the exact zeros), but 12.7% of nonzero returns still lie within 0.5% in magnitude – a dense cluster of small genuine adjustments immediately adjacent to the point mass at exactly zero, on top of the quiet/volatile regime structure the Markov-switching model already captures. The cedi’s return-generating process appears layered at least three deep – administrative holds, small routine adjustments, and rare large devaluations – a structure no single hurdle-plus-continuous architecture tested here fully resolves.

Appendix E. Pipeline validation on synthetic data

Before real data was available, the full pipeline was validated on a synthetic dataset engineered with known regime-dependent tail dependence, a simulated 2024-style cocoa shock, and a built-in cedi transmission channel. That run, independently re-executed as part of this paper’s verification process using the pipeline’s own fixed random seed, confirms the machinery recovers engineered structure: the Student- t copula is AIC-preferred for all ten pairs,

Table D.7: Cedi-pair significance (p -value, uncorrected, combined-stress, $q = 0.05$ lower tail) across all six marginal treatments.

Pair	GARCH	Raw ranks	ARMA-only	Weekly	Hurdle	Markov
Cocoa–Cedi	0.846	0.778	0.782	1.000	0.763	0.363
Gold–Cedi	0.036	0.148	0.136	1.000	0.132	0.100
Brent–Cedi	0.056	0.052	0.036	0.515	0.048	0.004
WTI–Cedi	0.012	0.008	0.012	0.611	0.008	0.048

mean empirical lower-tail dependence rises from 0.18 (calm) to 0.31 (stress), and five of ten pairs show a statistically significant *increase* in tail dependence during stress – the opposite qualitative pattern from the real-data results in Section 5, which is the point: the same pipeline, applied to data with an engineered contagion effect, detects it, which is what licenses trusting its failure to detect the same effect in the real data as a genuine finding rather than a limitation of the estimation code.

Appendix F. Decision log

This appendix documents, in the order discovered, the data-construction issues identified and resolved during this analysis.

Exports variable-filtering error. The underlying Bank of Ghana exports table stacks eight distinct series – total merchandise exports (f.o.b.), the trade balance (legitimately negative in deficit months), and six commodity sub-components – in long form under a single value column. An initial extraction read this column without filtering by the accompanying variable label, mixing negative trade-balance observations into what was intended as a pure export-level series; the symptom was a downstream export-growth regression returning an undefined R^2 , since the logarithm of a negative level is undefined. The fix filters strictly to the “Merchandise Exports (f.o.b)” label; re-derivation directly from the raw table confirms 228 monthly observations with zero remaining negative values.

Reporting-gap encoding. Independently, the same raw table encodes an eight-month reporting gap (May–December 2023) as a literal value of 0.00 rather than a missing-data marker. Ghana’s exports did not collapse to zero in mid-2023; these eight months are masked to missing rather than used, yielding 100 valid monthly observations within the 2015-01 to 2023-04 analysis window, cross-validated against the transmission regressions’ reported sample sizes in Section 5 ($108 - 1 - 11 = 96$ for the CPI regression and $100 - 1 - 11 = 88$ for the export-growth regression, both matching exactly once first-differencing and the daily-tail regressors’ burn-in period are accounted for).

Bootstrap resolution. Reported in Section 5.4: an initial 500-replicate bootstrap pass on the COVID-only disaggregated stress window produced six apparently Bonferroni-significant cells, all with p -values at the resolution floor a 500-replicate bootstrap can produce; re-estimation at 5,000 and 15,000 replicates resolved all six to non-significant.

Appendix G. Verification protocol

Every quantitative claim in this paper is checked against a specific source file and cell, row, or computation, recorded in a machine-readable ledger distributed with the code repository (Data Availability Statement). Three discrepancies were found and corrected between an earlier working draft and the underlying output tables during this process, the most consequential being an initially reversed description of the goodness-of-fit result in Section 5.3 (the draft stated no family was rejected for six named pairs; the source tables show the opposite, that every family is rejected for those six pairs) – corrected in the version reported here, which the source data support directly.